# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

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|---|---|---|---|
| **Course Code:** | ITA0502 | **Course Title:** | COMPUTER VISION |
| **CO Assessed:** | CO2 – Manipulation (BL3) | **Assessment:** | Assessment Tool 2 – Scenario Based Assessment |
| **Weightage:** | 20% of CIA | **Total Marks:** | 20 (2 Questions × 10 Marks) |
| **CO5 Statement:** | *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)* | | |
| **Student Name:** | Shalini Douglas | **Reg. No.:** | 192321019 |
| **Section / Batch:** | SLOT C / 2023 | **Date:** | |

**Code of Conduct**

*I **Shalini Douglas (192321019)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Shalini Douglas**

**Instructions**

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

| Section / Topic | Focus Area | Q. Nos. | Marks |
|---|---|---|---|
| Frequency Domain Image Enhancement | Periodic noise removal, Fourier Transform, notch filtering, quality inspection | 1 | 10 |
| Spatial Filtering & Edge-Preserving Enhancement | Bilateral filtering, edge preservation, autonomous vehicle navigation | 2 | 10 |
| | | **GRAND TOTAL:** | **20 Marks** |

## Annexure A – Scenario Based Assessment

### Scenario 1: Manufacturing Quality Inspection

**An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.**

**(a) Interpret the scenario and identify the type of noise present.**
**(b) Explain how this degradation affects inspection accuracy.**
**(c) Recommend an appropriate enhancement technique to remove the interference.**
**(d) Justify why your selected approach is more suitable than spatial-domain methods.**
**(e) Explain your answer in a logical sequence.**

### Scenario 2: Autonomous Vehicle Navigation

**A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.**

**(a) Interpret the scenario and explain the challenge involved.**
**(b) Identify the importance of preserving edges while removing noise.**
**(c) Recommend a suitable filtering technique.**
**(d) Justify why your selected filter preserves important image details.**
**(e) Present your explanation clearly and systematically.**

## Rubrics for Calculation

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) | Marks | |
|---|---|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario | | |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues | | |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts | | |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided | | |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand | | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| | | |
| Signature: _______________ | Signature: _______________ | Signature: _______________ |
| Date: ____________________ | Date: ____________________ | Date: ____________________ |

## Scenario 1: Manufacturing Quality Inspection

(a) Original Image

(b) Image with Periodic Nois

**(a) Interpret the scenario and identify the type of noise present.**

The inspection system captures images of metal parts to identify manufacturing defects. Due to electrical interference during image acquisition, regular horizontal lines appear across the images.

**Type of noise:**

- **Periodic noise (horizontal stripe noise)** caused by electrical interference.

**(b) Explain how this degradation affects inspection accuracy.**

The horizontal stripes reduce image clarity and interfere with the visibility of the component surface.

**Effects include:**

- Small defects such as scratches or cracks may be hidden.
- The inspection system may mistake the stripes for actual defects.
- Feature extraction becomes less accurate.
- Overall defect detection reliability decreases.

**(c) Recommend an appropriate enhancement technique to remove the interference.**
**Recommended technique:**

- Frequency-domain filtering using a Notch Reject Filter.

This method transforms the image into the frequency domain, removes the frequencies responsible for the stripe pattern, and reconstructs the enhanced image.

**(d) Justify why your selected approach is more suitable than spatial-domain methods.**
A notch filter is more effective because periodic noise appears at specific frequency locations.
**Advantages:**

- Removes only the unwanted frequency components.
- Preserves the important details of the metal surface.
- Avoids unnecessary image blurring.

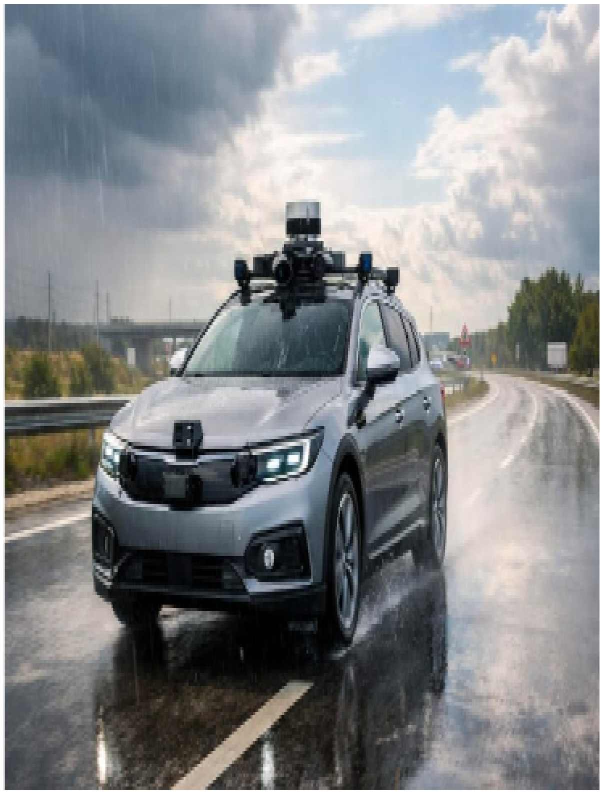
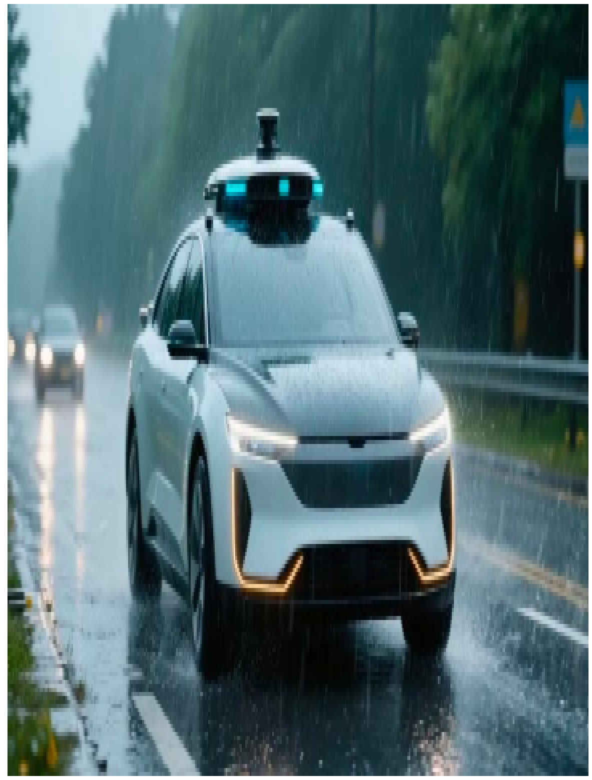
Spatial filters generally smooth the whole image and cannot selectively eliminate periodic stripe noise.

**(e) Explain your answer in a logical sequence.**
**Step-by-Step Process**

1. Capture the image of the metal component.
2. Horizontal stripe noise appears due to electrical interference.
3. Apply the Fourier Transform.
4. Identify the unwanted frequency components.
5. Remove them using a notch reject filter.
6. Perform the inverse Fourier Transform.
7. Obtain a clearer image for reliable defect inspection

**Conclusion**
Using a **frequency-domain notch filter** effectively removes periodic stripe noise while preserving important image details, leading to more accurate quality inspection.

**Scenario 2: Autonomous Vehicle Navigation**

**(a) Interpret the scenario and explain the challenge involved.**
The autonomous vehicle captures road images during heavy rainfall. Rain introduces noise into the images, making it difficult to clearly identify lane markings and road boundaries.
**Challenge:**
The image must be denoised without losing the sharpness of essential road features.

**(b) Identify the importance of preserving edges while removing noise.**
Edge preservation is necessary because edges define important road information.
Benefits include:

- Accurate lane detection.
- Clear identification of road boundaries.
- Better obstacle recognition.
- Improved navigation decisions.

If edges become blurred, the vehicle may incorrectly interpret the road.

**(c) Recommend a suitable filtering technique.**
**Recommended filter:**

- Bilateral Filter

The bilateral filter removes image noise while maintaining edge information.

**(d) Justify why your selected filter preserves important image details.**
The bilateral filter considers both the distance between pixels and their intensity differences.
**As a result:**

- Noise is reduced in smooth regions.
- Pixels across edges are not averaged together.
- Lane markings and road edges remain sharp.
- Important image structures are preserved.

**(e) Present your explanation clearly and systematically.**
**Step-by-Step Process**

1. Capture the road image during rainfall.
2. Rain introduces unwanted image noise.
3. Apply a bilateral filter.
4. Reduce noise while preserving edges.
5. Detect lanes and road boundaries.
6. Use the enhanced image for safe vehicle navigation.

**Conclusion**
The **bilateral filter** effectively removes rain noise while preserving important road edges, allowing the autonomous vehicle to detect lanes accurately and navigate safely.